Autonomy of Unmanned Aerial Vehicles

Practice 06 - Part B

by

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Tecnológico de Monterrey

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Academic Minor

Autonomy of Unmanned Aerial Vehicles

It is an advanced level course in control engineering, in which the student implements their own control algorithms, applying the competencies developed in the focus stage of their study plan.

- It requires prior knowledge of modeling and automation and control system design.
- The student supports his proposal by the analysis of the performance of the control system.

QDrone 2 | Qube-Servo 3 | Aero 2

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Configuration 1

Qube-Servo 3

Configuration 2

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Qube-Servo 3

It is supplied with two standard items:

- Inverted pendulum

  The inverted pendulum is mounted to the equipment with magnets. Here, the encoder provide an additional sensor input to the system [6].

- Inertia disk

  A small aluminum part is mounted to the equipment using magnets.

Fig. 1 Inverted pendulum.

Fig. 2 Inertia disk.

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Context

Abstract

Most PID controllers are adjusted on-site; herein, many different types of tuning rules have been proposed in the literature. The proportional term is based on the present error, the integral term depends on past errors, and the derivative term is a prediction of future errors. However, its implementation may not lead to a good system response for real-world systems. The main drawbacks of basic parallel PID controllers are the effects caused by the lack of proportional action on the process. In order to minimize these effects, modification is made to the parallel PID regulator structure. For this reason, in this lab practice, a new control strategy, called I+P control, is proposed to regulate the angular velocity of the QUBE-Servo 3. Hence, analysis of the control signal is performed and compared with the PI controller.

Topics Covered

- Velocity control
- Proportional-integral-derivative control
- Design control according to specifications

Prerequisites

- Hardware Interfacing laboratory experiment
- Matlab® script of 2-D line plot

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Introduction

The transfer function $P(s)$ that relates the angular position and the voltage of the system, $\Omega_m(s)/V_m(s)$ obtained in Lab Practice 3 is the following,

$$P(s) = \frac{k_t}{J_{eq}R_m s + k_t(k_m + \mu)} \tag{1.1}$$

Thus, a proportional, integral, and derivative (PID) control can be implemented to regulate the angular velocity of the QUBE-Servo 3. Here, the control law can be mathematically expressed as

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{1.2}$$

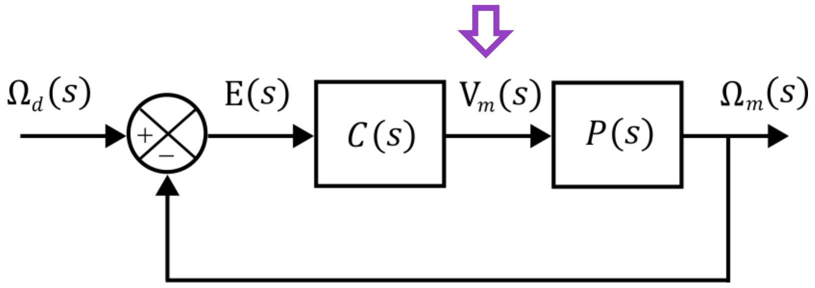

Fig. 1.1 Feedback control loop.

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Introduction

Henceforth, based on Eq. 1.2 and Fig. 1.1, the block diagram of the feedback loop under the PID control strategy is presented in Fig. 1.2

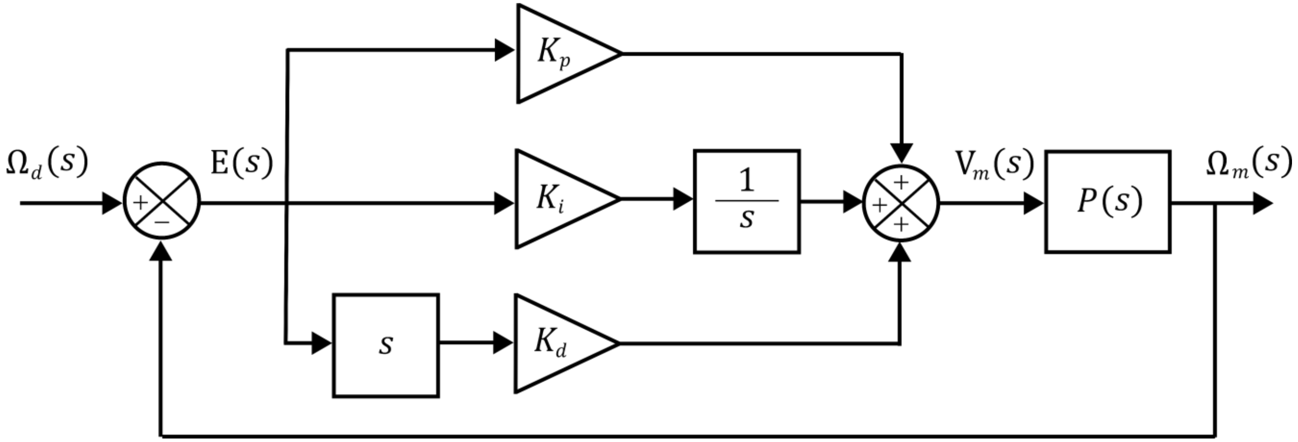

Fig. 1.2 Block diagram of PID control.

Remark 1: The PID controller may not lead to a good system response for real-world systems.

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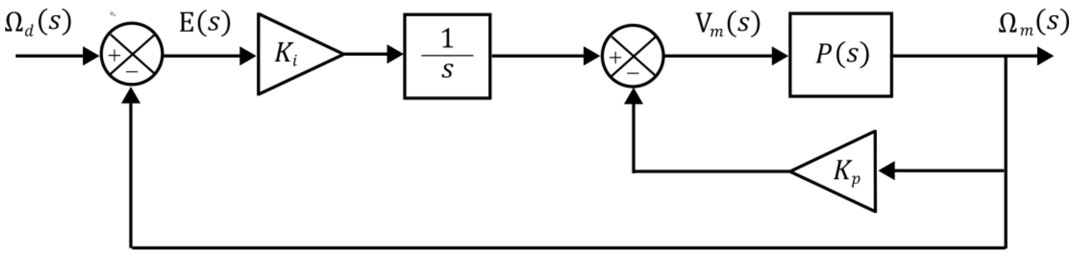
Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.



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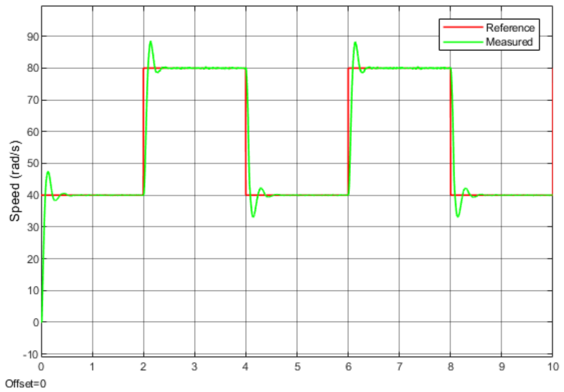
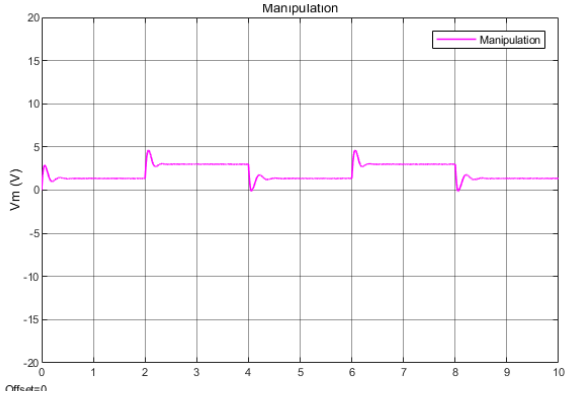
Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design the model of integral plus proportional (I+P) feedback control
3. Plot results ($K_i = 2.5[\mathrm{V/(rad-s)}]$, $K_p = 0.05[\mathrm{V-s/rad}]$).





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Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design the model of integral plus proportional (I+P) feedback control
3. Plot results ($K_i = 2.5[\mathrm{V/(rad-s)}], K_p = 0.05[\mathrm{V-s/rad}]$).
4. Perform a qualitative analysis of the QUARC® controller.

   i. Vary $K_i = 1 - 4\ [\mathrm{V/(rad-s)}]$ keep $K_p = 0\ [\mathrm{V-s/rad}]$

   ii. Keep $K_i = 2.5\ [\mathrm{V/(rad-s)}]$ vary $K_p = 0 - 0.15\ [\mathrm{V-s/rad}]$

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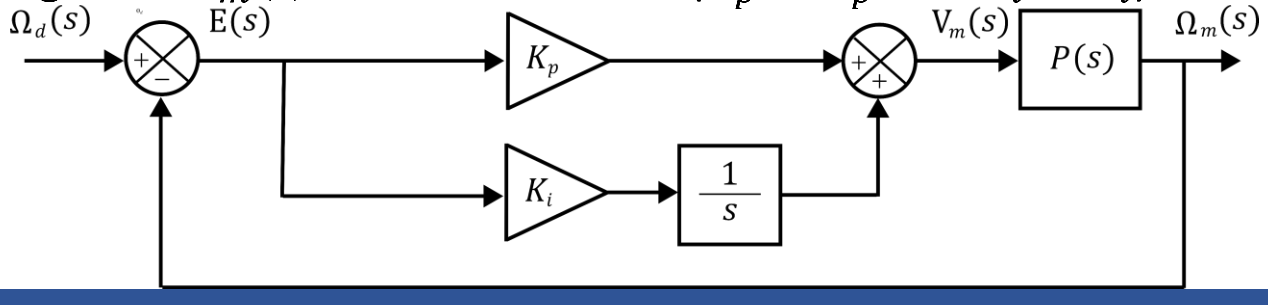
Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design the model of integral plus proportional (I+P) feedback control
3. Plot results ($K_i = 2.5[\mathrm{V}/(\mathrm{rad}-\mathrm{s})]$, $K_p = 0.05[\mathrm{V}-\mathrm{s}/\mathrm{rad}]$).
4. Perform a qualitative analysis of the QUARC® controller.
5. Obtain the closed-loop transfer function $\Omega_m(s)/\Omega_d(s)$.
6. Find K_i and K_p to achieve $\%MP = 2.5\ \%$ and $t_p = 0.15\ [\mathrm{sec}]$ in $\omega_m(t)$
7. Regulate $\omega_m(t)$ with a PI control ($K_p = K_p$ and $K_i = K_i$).



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8. Conclusions.

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Thank you!

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